Singly Linked List- Application

Write a C program to implement a **Singly Linked List** to store student records consisting of roll number, name, and marks. The program should allow the user to create a list of *n* students and perform various operations such as inserting a new student record at the beginning and at the end of the list, as well as inserting a record after the first student having an even roll number. Additionally, the program should display all student records, display only those students with even roll numbers, find and display the student who has obtained the maximum marks, and search for a student using their roll number. The program must be menu-driven and should use dynamic memory allocation for managing the linked list.

```
#include<stdio.h>
#include<stdlib.h>

struct student
{
    int roll, marks;
    char name[20];
    struct student *next;
};

struct student *create(int n)
{
    struct student *head=NULL,*temp=NULL,*newnode;
    int i;

    for(i=1;i<=n;i++)
    {
        newnode=(struct student*)malloc(sizeof(struct
student));

        printf("\nEnter Roll No: ");
        scanf("%d",&newnode->roll);

        printf("Enter Name: ");
        scanf("%s",newnode->name);

        printf("Enter Marks: ");
        scanf("%d",&newnode->marks);

        newnode->next=NULL;

        if(head==NULL)
        {
            head=newnode;
            temp=newnode;
        }
        else
        {
            temp->next=newnode;
            temp=newnode;
        }
    }

    return head;
}
```

```
void display(struct student *head)
{
    struct student *temp=head;

    while(temp!=NULL)
    {
        printf("\nRoll:%d Name:%s Marks:%d",
        temp->roll,temp->name,temp->marks);

        temp=temp->next;
    }
}

struct student* insert_begin(struct student *head)
{
    struct student *newnode;

    newnode=(struct student*)malloc(sizeof(struct
student));

    printf("Enter Roll No: ");
    scanf("%d",&newnode->roll);

    printf("Enter Name: ");
    scanf("%s",newnode->name);

    printf("Enter Marks: ");
    scanf("%d",&newnode->marks);

    newnode->next=head;
    head=newnode;

    return head;
}

struct student* insert_end(struct student *head)
{
    struct student *newnode,*temp=head;

    newnode=(struct student*)malloc(sizeof(struct
student));

    printf("Enter Roll No: ");
    scanf("%d",&newnode->roll);

    printf("Enter Name: ");
    scanf("%s",newnode->name);
```

```
    printf("Enter Marks: ");
    scanf("%d",&newnode->marks);

    newnode->next=NULL;

    if(head==NULL)
    {
        head=newnode;
        return head;
    }

    while(temp->next!=NULL)
        temp=temp->next;

    temp->next=newnode;

    return head;
}

struct student* insert_after_even(struct student *head)
{
    struct student *temp=head,*newnode;

    while(temp!=NULL)
    {
        if(temp->roll%2==0)
        {
            newnode=(struct student*)malloc(sizeof(struct
student));

            printf("Enter Roll No: ");
            scanf("%d",&newnode->roll);

            printf("Enter Name: ");
            scanf("%s",newnode->name);

            printf("Enter Marks: ");
            scanf("%d",&newnode->marks);

            newnode->next=temp->next;
            temp->next=newnode;

            return head;
        }

        temp=temp->next;
```

```
    }

    printf("No even roll number found");
    return head;
}

void even_roll(struct student *head)
{
    struct student *temp=head;

    printf("\nEven Roll Number Students\n");

    while(temp!=NULL)
    {
        if(temp->roll%2==0)
        printf("\nRoll:%d Name:%s Marks:%d",
        temp->roll,temp->name,temp->marks);

        temp=temp->next;
    }
}

void max_marks(struct student *head)
{
    struct student *temp=head,*max=head;

    if(head==NULL)
    {
        printf("List Empty");
        return;
    }

    while(temp!=NULL)
    {
        if(temp->marks>max->marks)
        max=temp;

        temp=temp->next;
    }

    printf("\nStudent with Max Marks");
    printf("\nRoll:%d Name:%s Marks:%d",
    max->roll,max->name,max->marks);
}
```

```
void search(struct student *head)
{
    int r;
    struct student *temp=head;

    printf("Enter roll number to search: ");
    scanf("%d",&r);

    while(temp!=NULL)
    {
        if(temp->roll==r)
        {
            printf("\nRecord Found");
            printf("\nRoll:%d Name:%s Marks:%d",
            temp->roll,temp->name,temp->marks);
            return;
        }

        temp=temp->next;
    }

    printf("Record not found");
}

int main()
{
    struct student *head=NULL;
    int n,ch;

    printf("Enter number of students: ");
    scanf("%d",&n);

    head=create(n);

    while(1)
    {
        printf("\n1.Insert Begin");
        printf("\n2.Insert End");
        printf("\n3.Insert After Even Roll");
        printf("\n4.Display");
        printf("\n5.Display Even Roll Students");
        printf("\n6.Max Marks");
        printf("\n7.Search Record");
        printf("\n8.Exit");

        printf("\nEnter choice: ");
        scanf("%d",&ch);
```

```
        switch(ch)
        {
            case 1: head=insert_begin(head); break;
            case 2: head=insert_end(head); break;
            case 3: head=insert_after_even(head); break;
            case 4: display(head); break;
            case 5: even_roll(head); break;
            case 6: max_marks(head); break;
            case 7: search(head); break;
            case 8: exit(0);
            default: printf("Invalid choice");
        }
    }
}
```